

EXPERIMENT 5

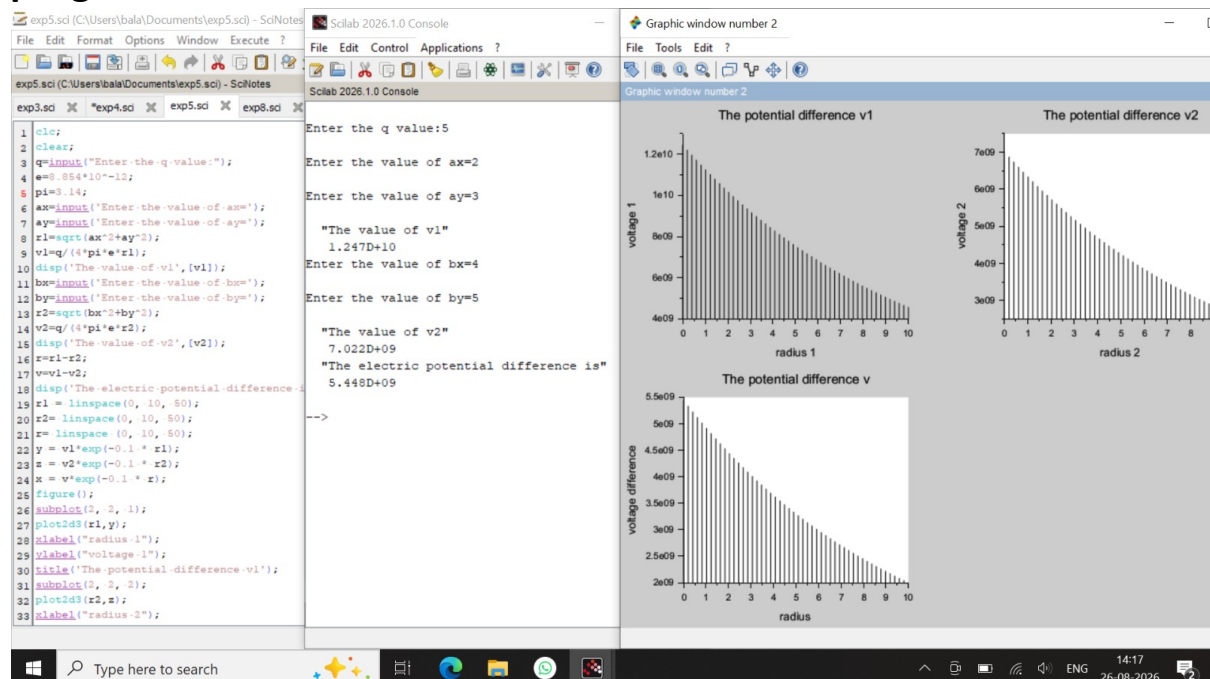
Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R.

Aim:

To write program for Electric potential difference between two points in free space medium using SCILAB. **Software required:**

•SCILAB version 6.0.1 **Code:**

program:



Theoretical Calculation:

Exp: 5: Electric potential at any two points

Manual cal:-

Calculate r_1
 $a_x = 2, a_y = 3$
 $r_1 = \sqrt{a_x^2 + a_y^2}$
 $r_1 = \sqrt{2^2 + 3^2}$
 $= \sqrt{4+9} = \sqrt{13} = 3.60555 \text{ m}$

Calculate V_1 :

$q = 5 \text{ C}$
 $\epsilon_0 = 8.854 \times 10^{-12}$
 $V_1 = \frac{q}{4\pi\epsilon_0 r_1}$
 $V_1 = \frac{5}{4(3.14)(8.854 \times 10^{-12})(3.60555)}$
 $V_1 = 1.2477 \times 10^{10} \text{ V} = 1247.08 \text{ V}$

Calculate r_2 :

$r_2 = \sqrt{b_x^2 + b_y^2}$
 $= \sqrt{4^2 + 5^2}$
 $= \sqrt{16+25} = \sqrt{41} = 6.40312 \text{ m}$

Calculate V_2

$V_2 = \frac{q}{4\pi\epsilon_0 r_2}$
 $V_2 = \frac{5}{4(3.14)(8.854 \times 10^{-12})(6.40312)}$
 $V_2 = 7.022 \times 10^9 \text{ V} \Rightarrow 7021.81 \text{ V}$

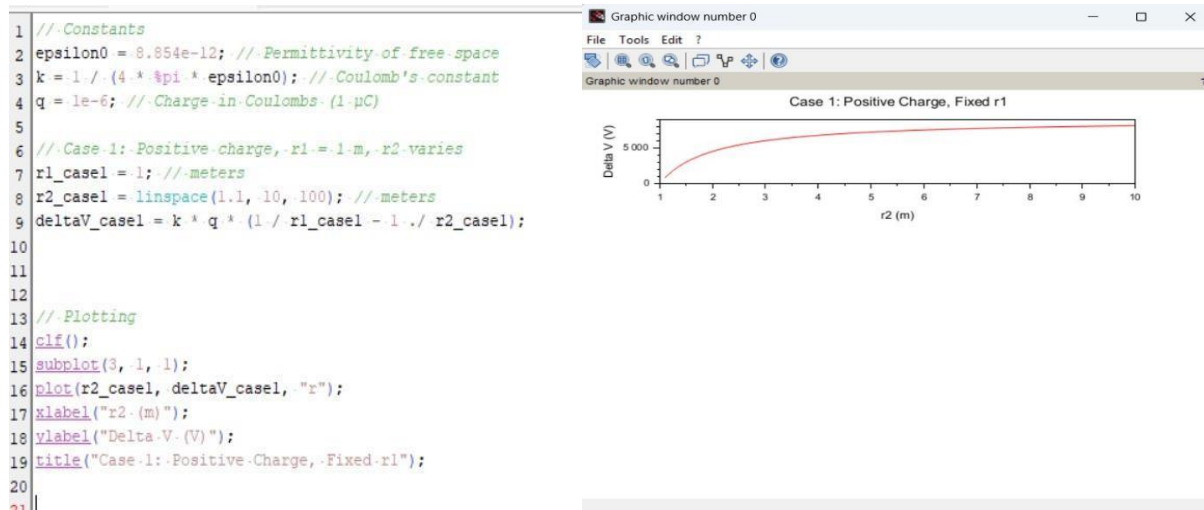
Potential Difference

$\Delta V = V_1 - V_2$
 $\Delta V = 1.2477 \times 10^{10} - 7.022 \times 10^9$
 $\Delta V = 5.448 \times 10^9 \text{ V}$

Case 1: A positive charge with fixed $r_1=1 \text{ m}$ r_2 varying.

Code:

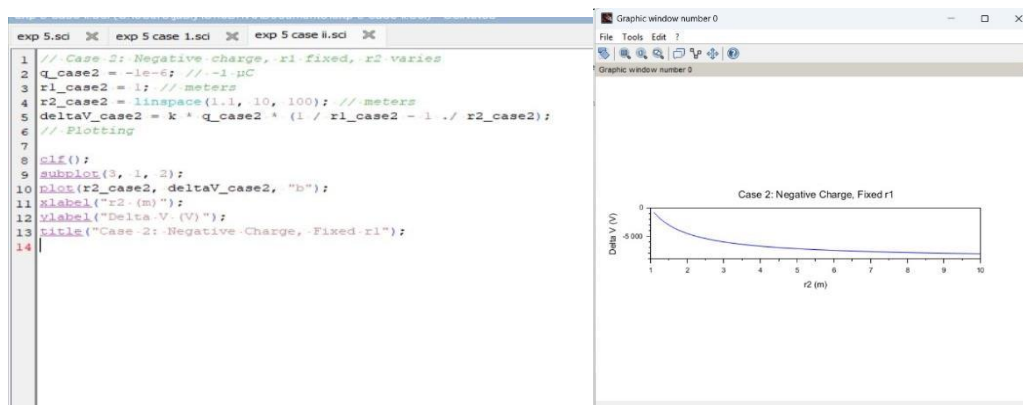
output:



Case 2: A negative charge with r_1 fixed and r_2 varying.

Code:

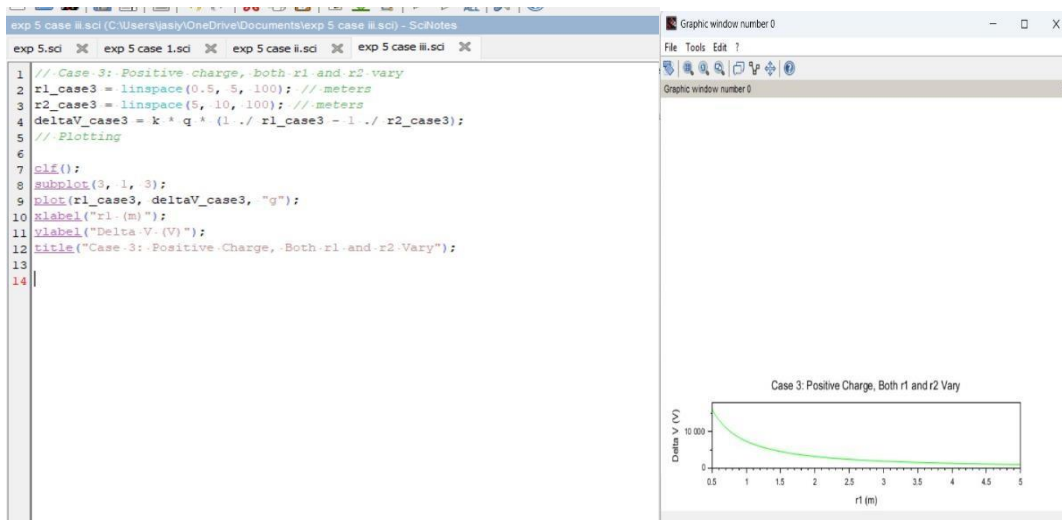
output:



Case 3: Both points r_1 and r_2 vary with a positive charge.

Code:

output:



Result :

Thus the program was verified for Electric potential difference between two points in free space medium using SCILAB was successfully.